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Control of diarrheal disease (CDD)

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Objectives

- To understand the what is diarrhea and types of diarrhea.
- To understand why management of diarrheal cases are crucial.
- To learn distinguish types of dehydration.



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CDD

Diarrhea:- is the passage of liquid or watery stool at least 3 times during 24 hr.

Consistency is more important than the frequency.

Usually in breast fed infant pass semi solid, pasty & yellow stool.

Sometimes they pass stool after each breast feeding ,this isn't diarrhea.



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Types of Diarrhea

Diarrhea can be:

- Acute which lasts for not more than 14 days.
- Persistent which lasts for 14 days or longer.



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Dysentery

***Dysentery** : blood in the stool (amoebic or bacillary).

500,000,000 children under 5 years of age are the major victim of diarrhea & its estimated that under 5 yr old may develop 1-12 episodes of diarrhea per a year.

It will lead to growth failure & complications.
Annually 3,000,000 children die from diarrhea all over the world .This means 6 children per minute.



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Diarrhea with sever dehydration require admission to the hospital, need special care, drugs, fluids ,etc.. These are very coasty.

The prognosis is usually bad probably ending in death.



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***Causes of diarrhea:**

1-viruses

2-bacteria.

3- Protozoa.



***Contributing factors**

- Unclean water.
- Dirty hands when preparing meals(mothers) ,
 - or when eating meals in children.
- Food spoilt by high temperature .

Diarrhea leads to malnutrition .this manifested as growth retardation & under weight.
(~~25gm~~ loss of weight for each day of diarrhea).

250 gm



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Factors leading to growth deficit in a baby with diarrhea:

- Reduced food intake .
 - Reduced absorption of nutrient due to rapid
 - gut transit time.
 - Catabolic losses through vomiting & fever
 - leading to energy expenditure.
- This malnutrition will lead to reduce immunity leading to further episodes of diarrhea.



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Risky infants

1-Artificially fed infants

2-At the weaning age: due to inadequate feeding, contaminated food & low resistance to infection.

Diarrhea is a protective mechanism ,wash microorganisms & toxin. It's a self limiting ,not a killer. The killer is the dehydration. (fluid-electrolyte deficit).



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Dehydration

Definition:- is deficit in water & electrolytes
(Sodium, Potassium ,Chloride & bicarbonate)

resulting from losses in the stool, vomiting ,
urine , fever, sweating & breathing.



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Assessment of diarrhea in diarrhea cases for dehydration

Sign A B C

General condition of the child	Well, alert	Restless ,irritable	Lethargic ,unconscious
Eyes		Sunken	Sunken
Thirst	Drink normally	Eager to drink	Unable to drink
Skin pinch	Goes back quickly	Goes back slowly	Goes back very slowly
Classification	No dehydration	Some dehydration	Sever dehydration
Treatment	Plan A (home)	B (PHC)	C (Hospital)



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Treatment plan:

Plan A : to preventing dehydration from occurrence..

1. Give extra fluid :ORS (oral rehydration solution).

2. Continue feeding (breast or others).

3. Teach the mother;

- How to prepare & give ORS.

- Signs of dehydration & danger signs by showing her picture of the main signs of dehydration.



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Treatment plan:

Plan B: The aim is to correct dehydration.

- Give ORS in the health center. Child's weight(Kg)*75ml
- =volume given over 4^{hours} period. Assess every hour.
- Continue feeding or breast feeding .
- Give 100-200ml of clean water.
- Teach mother to prepare & give ORS correctly(cup & spoon) as in plan A. Assess every hour.



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- If the child vomits the ORS, wait for 10 minutes & then
- restart giving him the solution slowly.

after the whole ORS given

- Reassess after four hour ,classify according to the
 - hydration status & use the appropriate plan accordingly.
-
- **Puffiness of the face & eyes is a sign of over**
 - **hydration ,in that case give the fluid slowly.**



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Plan C: The aim is to correct the dehydration urgently intravenously or through N.G tube I.V: give Ringer lactate solution if it is not use N.S.
Weight (Kg) * 100ml over a period of 3hr for children over 1 year ,over 6 hr. for infant.

I.V. rehydration by age & time

Age(months)	Amount of I.V fluid/unit time	
	30ml/kg body weight	70ml/kg body
weight		

<12 <i>→ infant</i>	first hour	following 5 hr
12+ <i>→ >1 year</i>	first 30minte	following 2.5hr



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The steps are:

1. Reassess every hour ,if no improvement give fluid more rapidly .
2. if the patient can drink ,give ORS in 5ml/kg body weight/hr.
3. Reassess after completion ,classify according to the hydration status & choose the appropriate plan accordingly .



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Oral rehydration solution

Oral rehydration solution

Composition:

Na: 3.5 gm

NaHCO₃: 2.2gm

KCL: 1.5gm

Glucose: 20gm

Dilute in 1000ml(1Lt) of water.

Advantage: Cheap, effective & easy to give.



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Diarrhea case fatality rate has decreased a lot after the introduction of ORS , due to prevention of dehydration.

Management of chronic (persistent) diarrhea.

IF the child < 6 months of age & dehydrated refer to hospital.



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If the child over 6 months & not dehydrated ,
management is mainly dietary.

Teach the mother :

1. Dilute any animal milk given to the child with
an equal volume of water or replace with
fermented milk products such as yogurt.



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continued:

2.Increases energy intake : 6 meal per day of thick cereals , added oils or fats , vegetables , pulses fish & meat.

3.Reassess in 5 days :If diarrhea persist refer to hospital .

If diarrhea has stopped teach the mother to use the regular diet ,re-asume the usual animal milk & give an extra meal every day for one month.



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Conclusion

- Distinguishing different types of diarrhea.
- Be able to manage diarrheal cases.
- Avoiding the dehydration from occurrence.



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Thank You

 1. Which of the following best defines diarrhea?

- A. Passage of formed stool three times daily
- B. Passage of watery or loose stool at least three times in 24 hours
- C. Passage of mucus in stool once daily
- D. Any stool passed after feeding

Answer:

 B. Passage of watery or loose stool at least three times in 24 hours

 2. What is more important than frequency when diagnosing diarrhea?

- A. Smell
- B. Color
- C. Consistency
- D. Amount

Answer:

 C. Consistency

 3. Which of the following is not a common cause of diarrhea?

- A. Viruses
- B. Bacteria
- C. Protozoa
- D. Fungi

Answer:

 D. Fungi

🩸 4. Dysentery is defined as diarrhea with:

- A. Fever
- B. Blood and mucus in stool
- C. Watery stool
- D. Vomiting

Answer:

➡ B. Blood and mucus in stool

👧 5. The most dangerous complication of diarrhea is:

- A. Weight loss
- B. Growth retardation
- C. Dehydration
- D. Fever

Answer:

➡ C. Dehydration

🍼 6. At what age does weaning commonly occur, increasing the risk of diarrhea?

- A. 0-2 months
- B. 3-5 months
- C. Around 6 months
- D. 1 year

Answer:

➡ C. Around 6 months

⚡ 7. A child presents with watery stool, sunken eyes, restless behavior, and slow skin pinch return. What is the dehydration level?

- A. No dehydration
- B. Some dehydration
- C. Severe dehydration
- D. Overhydration

Answer:

 B. Some dehydration

 8. In severe dehydration, the preferred IV fluid is:

- A. 5% dextrose
- B. Ringer Lactate
- C. Normal saline
- D. Half-normal saline

Answer:

 B. Ringer Lactate

 9. In Plan B (some dehydration), how much ORS should be given?

- A. 50 ml/kg over 2 hours
- B. 75 ml/kg over 4 hours
- C. 100 ml/kg over 6 hours
- D. 125 ml/kg over 3 hours

Answer:

 B. 75 ml/kg over 4 hours

 10. For Plan C, a 10-kg infant should receive how much IV fluid total?

- A. 300 ml
- B. 700 ml
- C. 1000 ml

D. 1500 ml

Answer:

 C. 1000 ml

$(100 \text{ ml/kg} \times 10 \text{ kg} = 1000 \text{ ml})$

 11. What is the main advantage of ORS?

- A. It replaces protein
- B. It's expensive but effective
- C. It prevents dehydration and is cheap
- D. It causes overhydration

Answer:

 C. It prevents dehydration and is cheap

 12. In a child >6 months with persistent diarrhea but no dehydration, the first step is:

- A. Start antibiotics
- B. Give anti-motility drugs
- C. Begin dietary management
- D. Stop all milk

Answer:

 C. Begin dietary management

 13. During diarrhea, animal milk should be:

- A. Completely stopped
- B. Diluted with equal water
- C. Given undiluted
- D. Replaced by plain water

Answer:

➡ B. Diluted with equal water

↺ 14. Puffiness of face and eyes during rehydration indicates:

- A. Underhydration
- B. Proper hydration
- C. Overhydration
- D. Dehydration

Answer:

➡ C. Overhydration



15. A mother should be taught to return if the child shows which of the following?

- A. Improved appetite
- B. Less thirst
- C. Increased sunken eyes and dry mouth
- D. Normal urine output

Answer:

➡ C. Increased sunken eyes and dry mouth

Perfect 👍 — now let's move from basic recall to clinical-style MCQs (16–30) where you apply what you learned. Each question gives a short case. Try to reason through it before checking the answer below.



16.

A 1-year-old boy passes watery stool 6 times a day for 2 days. He's alert, drinks eagerly, has sunken eyes, and his skin pinch returns slowly.

What is the level of dehydration?

- A. No dehydration
- B. Some dehydration
- C. Severe dehydration
- D. Overhydration

Answer:

 B. Some dehydration

 17.

The same child (above) weighs 8 kg. How much ORS should be given in Plan B?

- A. 400 ml over 4 hr
- B. 600 ml over 4 hr
- C. 800 ml over 4 hr
- D. 1000 ml over 4 hr

Answer:

 B. 600 ml

$(8 \text{ kg} \times 75 \text{ ml} = 600 \text{ ml total over 4 hours})$

 18.

A 3-year-old with severe dehydration is started on IV therapy. How much total fluid should he get?

- A. 50 ml/kg
- B. 75 ml/kg
- C. 100 ml/kg
- D. 125 ml/kg

Answer:

 C. 100 ml/kg

 19.

A 10-kg child (≥ 12 months) is given IV fluid for severe dehydration. What is the correct schedule?

- A. 30 ml/kg in 1 hr, 70 ml/kg in 5 hr
- B. 30 ml/kg in 30 min, 70 ml/kg in 2.5 hr
- C. 50 ml/kg in 1 hr, 50 ml/kg in 3 hr
- D. 100 ml/kg in 6 hr

Answer:

 B. 30 ml/kg in 30 min, 70 ml/kg in 2.5 hr

 20.

A 5-month-old baby with diarrhea and dehydration comes to the clinic. What should you do first?

- A. Manage at home
- B. Start ORS and reassess after 4 hours
- C. Admit to hospital
- D. Give antibiotics

Answer:

 C. Admit to hospital

(Infants < 6 months with dehydration need hospital care.)



21.

A 9-month-old has persistent diarrhea (> 14 days) but no dehydration. What is the key management step?

- A. Stop feeding until diarrhea stops
- B. Give diluted animal milk or yogurt and increase meals
- C. Give antibiotics
- D. Restrict fluids

Answer:

➡ B. Give diluted animal milk or yogurt and increase meals



22.

A 4-year-old presents with fever, abdominal cramps, and small stools containing blood and mucus. The most likely diagnosis is:

- A. Acute watery diarrhea
- B. Dysentery
- C. Cholera
- D. Persistent diarrhea

Answer:

➡ B. Dysentery



23.

Which of the following is the best initial fluid for IV rehydration in severe diarrhea?

- A. 5% Dextrose
- B. Ringer Lactate
- C. Half-normal saline
- D. Tap water

Answer:

➡ B. Ringer Lactate



24.

Why is dehydration the main killer in diarrhea?

- A. Because of toxins in stool
- B. Because the intestines stop functioning
- C. Because of fluid and electrolyte loss leading to shock
- D. Because of fever

Answer:

➡ C. Fluid and electrolyte loss leading to shock



25.

A mother reports her breastfed baby passes soft, yellow stool after each feed. What should you tell her?

- A. The baby has diarrhea
- B. It's normal; continue breastfeeding

- C. Stop breastfeeding for 24 hours
- D. Give ORS at home

Answer:

 B. It's normal; continue breastfeeding

 26.

Which of the following indicates severe dehydration?

- A. Restless child, drinks eagerly
- B. Sunken eyes, drinks normally
- C. Lethargic/unconscious, skin pinch goes back very slowly
- D. Normal skin pinch and normal thirst

Answer:

 C. Lethargic/unconscious, skin pinch goes back very slowly

 27.

A child with diarrhea develops puffy eyes during ORS therapy. What should be done?

- A. Stop fluids immediately
- B. Slow down fluid administration
- C. Increase fluid rate
- D. Give diuretics

Answer:

 B. Slow down fluid administration
(Puffiness = overhydration)

 28.

During counseling, the mother of a dehydrated child should be taught all except:

- A. How to prepare ORS
- B. Signs of dehydration
- C. How to mix antibiotics at home
- D. To continue breastfeeding

Answer:

 C. How to mix antibiotics at home

 29.

After diarrhea stops, the mother should:

- A. Stop feeding for two days
- B. Give normal diet plus one extra meal daily for a month
- C. Continue diluted milk forever
- D. Start antibiotics

Answer:

 B. Give normal diet plus one extra meal daily for a month

30.

Which component in ORS helps glucose-sodium co-transport for water absorption?

- A. Potassium chloride
- B. Sodium bicarbonate
- C. Glucose
- D. Water

Answer:



C. Glucose
